Java Web App Deployment on AWS EC2 + MySQL (Final Corrected)

1. Lab Objective

- Launch 2 EC2 instances
- Configure MySQL database
- Deploy Java Web App on Tomcat
- Connect App VM → DB VM
- Verify registration & login

2. Architecture Overview

App Server: jitendra-appserver

DB Server: jitendra-databaseserver

Flow:

Browser → App VM (8080) → DB VM (3306)

3. Launch EC2 Instances

DB VM:

- Name: jitendra-databaseserver
- Security Group:

  SSH (22) → My IP

  MySQL (3306) → sg-app-vm

App VM:

- Name: jitendra-appserver

- Security Group:

  SSH (22) → My IP

  HTTP (8080) → Anywhere

4. Configure DB VM

Install MySQL:

```
sudo apt update -y
sudo apt install mysql-server -y
sudo systemctl start mysql
sudo systemctl enable mysql
```

Create Database:

```
CREATE DATABASE jet;
USE jet;
```

Create Table:

```
CREATE TABLE USER (
id INT AUTO_INCREMENT PRIMARY KEY,
first_name VARCHAR(50),
last_name VARCHAR(50),
email VARCHAR(100),
username VARCHAR(50),
password VARCHAR(50),
```

```
regdate DATE
);
```

Create User:

```
CREATE USER 'appuser'@'%' IDENTIFIED BY 'YourPassword123!';
GRANT ALL PRIVILEGES ON jet.* TO 'appuser'@'%';
FLUSH PRIVILEGES;
```

Enable Remote Access:

```
sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf
```

Change:

```
bind-address = 127.0.0.1
```

To:

```
bind-address = 0.0.0.0
```

Restart MySQL:

```
sudo systemctl restart mysql
```

Verify:

```
sudo ss -tlnp | grep 3306
```

5. Configure App VM

Install Java & Maven:

```
sudo apt install openjdk-11-jdk -y

sudo apt install maven -y
```

Install Tomcat:

```
cd /opt

wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.91/bin/apache-tomcat-9.0.91.tar.gz

tar -xvf apache-tomcat-9.0.91.tar.gz

mv apache-tomcat-9.0.91 tomcat9

chmod +x /opt/tomcat9/bin/*.sh
```

Fix Permissions:

```
sudo chown -R ubuntu:ubuntu /opt/tomcat9
```

Start Tomcat:

```
sudo /opt/tomcat9/bin/startup.sh
```

Clone Project:

```
git clone https://github.com/Akiranred/aws-rds-java.git

cd aws-rds-java
```

Update DB Connection in JSP:

```
Class.forName("com.mysql.cj.jdbc.Driver");


Connection con = DriverManager.getConnection(
```

"jdbc:mysql://<DB-PRIVATE-IP>:3306/jet?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC",

"appuser",

"YourPassword123!"

);

Build Project:

mvn clean package

Deploy:

sudo /opt/tomcat9/bin/shutdown.sh

sudo rm -rf /opt/tomcat9/webapps/LoginWebApp*

sudo cp target/LoginWebApp.war /opt/tomcat9/webapps/

sudo /opt/tomcat9/bin/startup.sh

Test Application:

http://<APP-IP>:8080/LoginWebApp

Verify Data:

mysql -u appuser -p -e "SELECT * FROM jet.USER;"

Summary:

- App deployed successfully

- DB connected

- Data stored and retrieved